Systems Engineering - Midterm Review

Team 9 - Semantic Transmitter - "Semantic Audio Communication using GenAI"

André Jacinto, João Abade, Carlos Sousa, José Santos, Eduardo Medeiros, Nuno Vieira, Guilherme Ferreira, Rodrigo Resende, Inês Faleiro, Teresinha Teixeira, Tomás Sousa

Faculdade de Engenharia - Universidade do Porto

April 22, 2026

1. Introduction
   - Team Organization
   - Challenge Description
2. Validated Requirements
3. Outcome of the Market Survey
4. System Breakdown Structure
5. Functional Architecture
6. Measures of Effectiveness and Performance
7. System Concept
8. Work Breakdown Structure + Gantt
9. Risk Management
10. Conclusion
    - Internal Evaluation
    - Next Steps

Presenters:

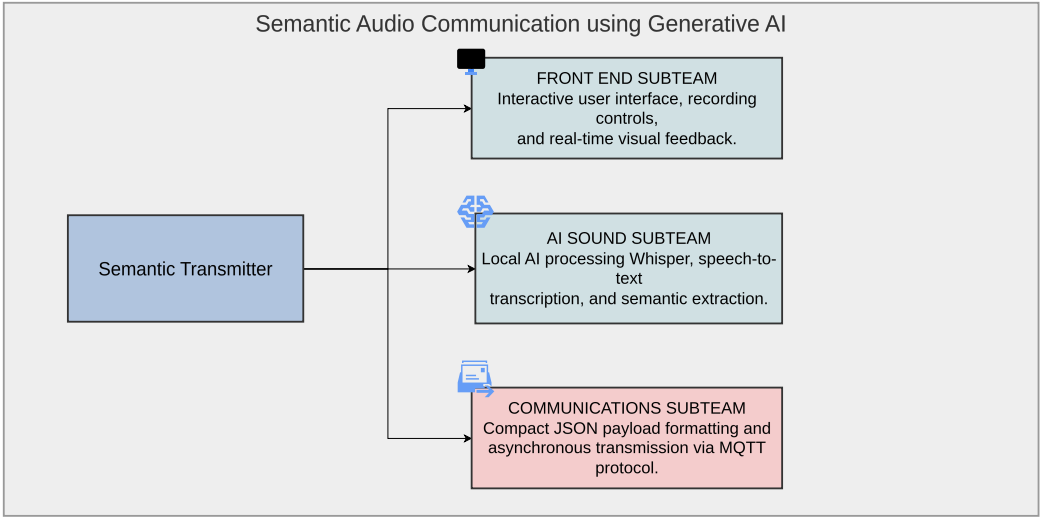
- João Abade
- Guilherme Ferreira
- Nuno Vieira
- José Santos
- Eduardo Medeiros

Supervisor:

- Professor Rui Lopes Campos

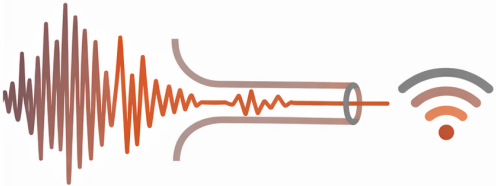
U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

Team organization



Challenge Description

The Problem (Status Quo)



Header: Traditional Voice Messaging

* High bandwidth consumption.
* Severe delivery delays on unstable links.

The Challenge/Our Solution

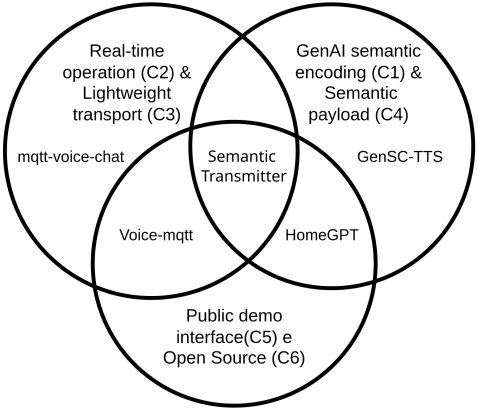
Header: Semantic Transmission

* Edge AI local meaning extraction.
* Smaller payload containing semantic information.
* Capable of working in unstable network conditions.

Validated Requirements

Category	Validated requirements
Communication	Asynchronous message sending
	Standardized semantic payload with transcript and metadata
	Delivery-state feedback
	Measurable bandwidth reduction vs. raw audio transmission
User Interface	Explicit start/stop recording controls
	Dynamic transcript display after processing
	Real-time status feedback
	Graceful, non-disruptive error handling during demos
AI STT Model	Model runs locally; retry path for capture/transcription failures; transcript shown before transmission; raw audio discarded after transcription; low end-to-end latency
Validation concept	Peer Reviews & Checklist Comparison

Outcome of the Market Survey



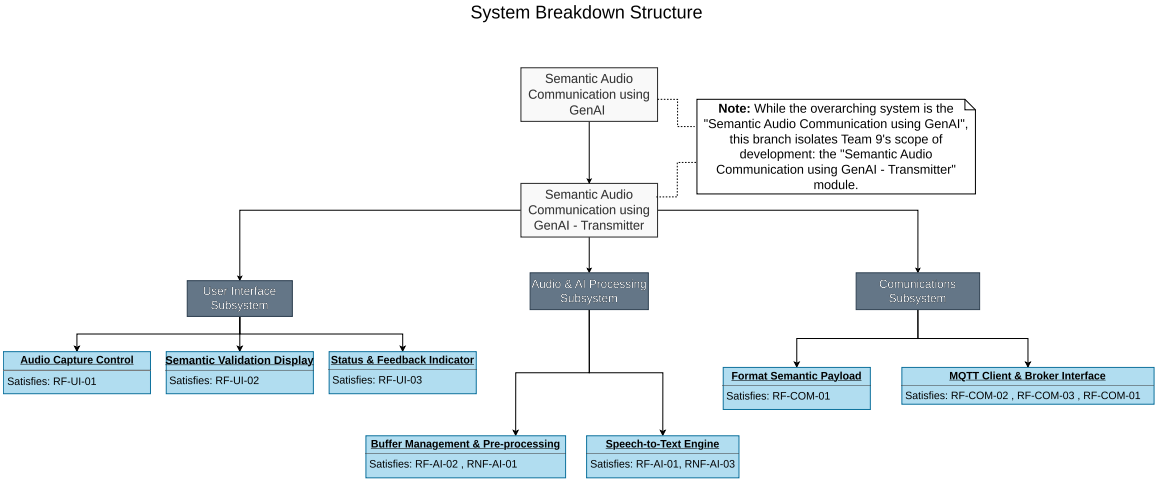
AI PROCESSING: Whisper large-v3

- High Accuracy STT & Native Portuguese Support.
- 100% Offline Capable (Critical for live demos)
- Free & Open-Source

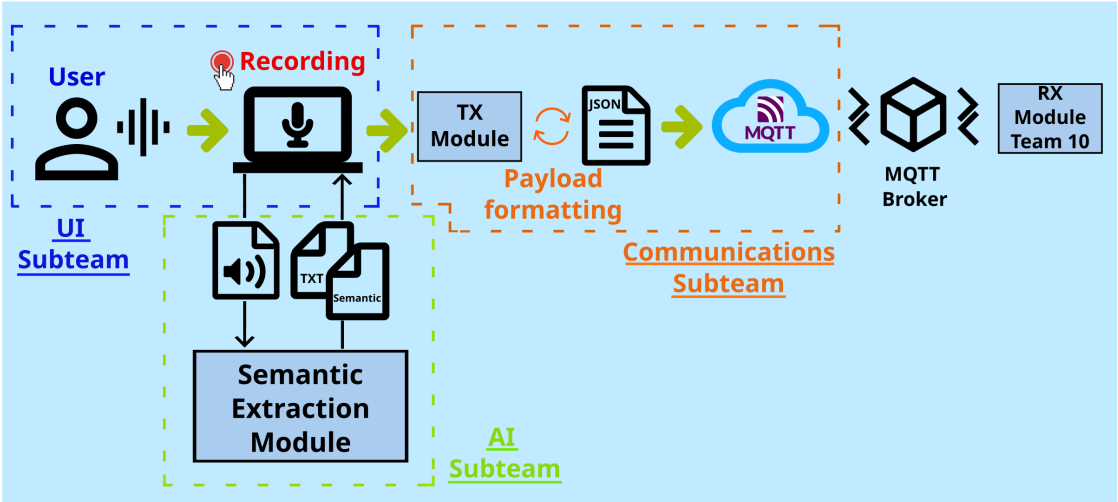
COMMUNICATION TRANSPORT: MQTT

- Minimal network overhead & very low latency
- Pub/Sub Broker Paradigm
- Asynchronous delivery (Decouples TX from RX)

System Breakdown Structure



Functional Architecture



Measures of Effectiveness and Performance

WER

Semantic Accuracy

- Evaluates the system's effectiveness in capturing the users's correct intent.
- If the WER is high, the main objective semantic communication) fails.

Delivery Reliability

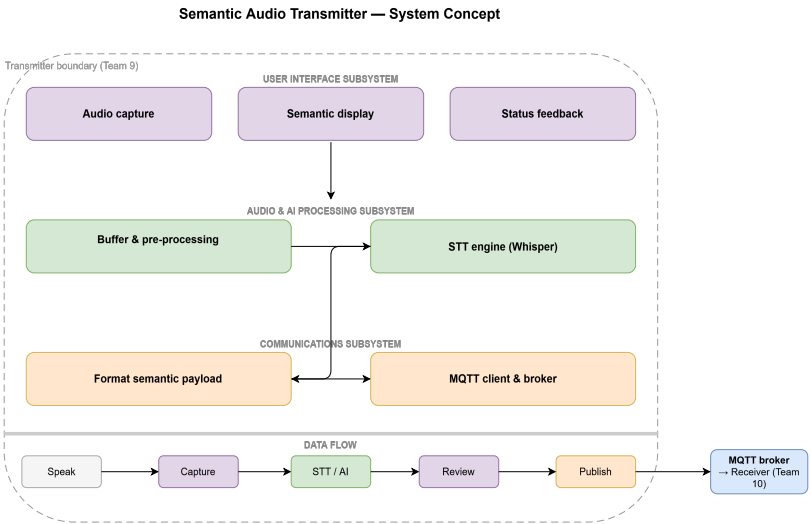
- The percentage of payload that are successfully published to the Broker out of the total number of attempts.
- The percentage of payloads that are compliant with the schema chosen to format the payload (JSON).

End-to-End Latency

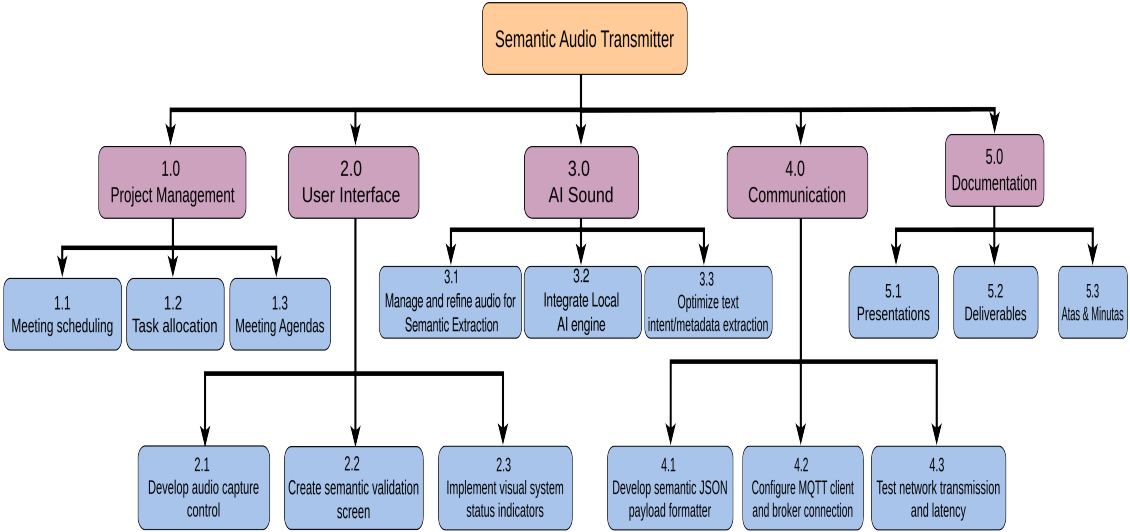
[MQTT Delivery Reliability]

- The total time measured from the moment the user finishes speaking until the formatted payload is sent.
- The risk report stipulates that the model's latency cannot exceed the defined limit.

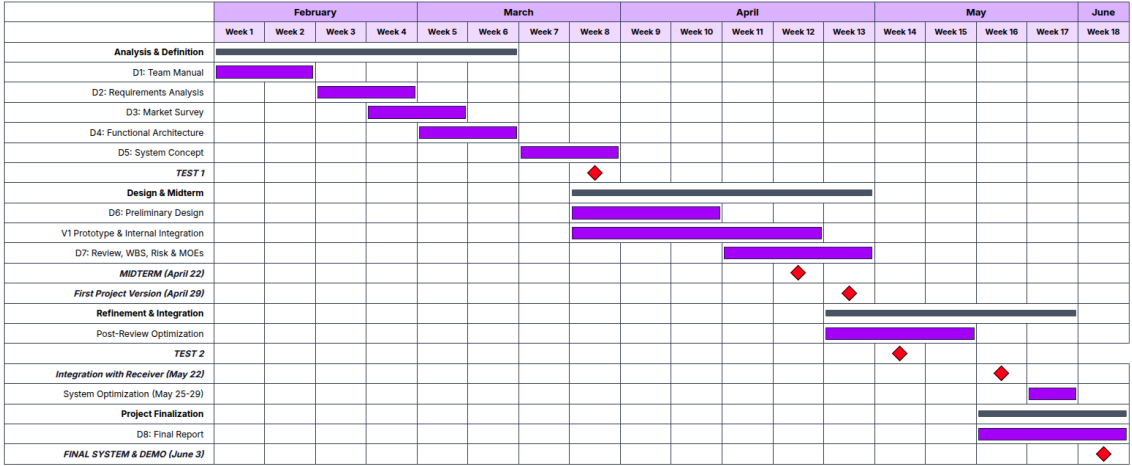
System Concept



Work Breakdown Structure + Gantt



Gantt Chart

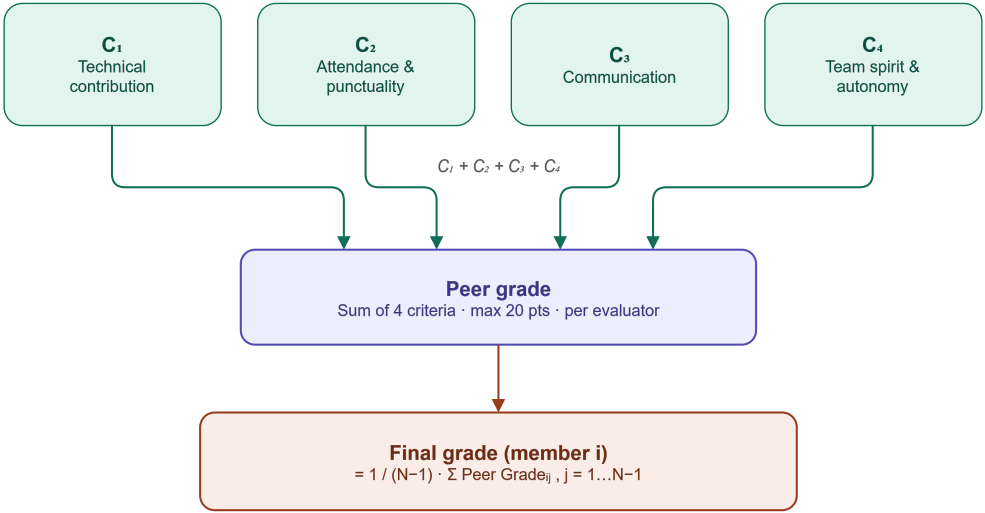

February
March
April
May
June
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
Week 13
Week 14
Week 15
Week 16
Week 17
Week 18
Analysis & Definition
D1: Team Manual
D2: Requirements Analysis
D3: Market Survey
D4: Functional Architecture
D5: System Concept
TEST 1
Design & Midterm
D6: Preliminary Design
V1 Prototype & Internal Integration
D7: Review, WBS, Risk & MOEs
MIDTERM (April 22)
First Project Version (April 29)
Refinement & Integration
Post-Review Optimization
TEST 2
Integration with Receiver (May 22)
System Optimization (May 25-29)
Project Finalization
D8: Final Report
FINAL SYSTEM & DEMO (June 3)

Risk Management

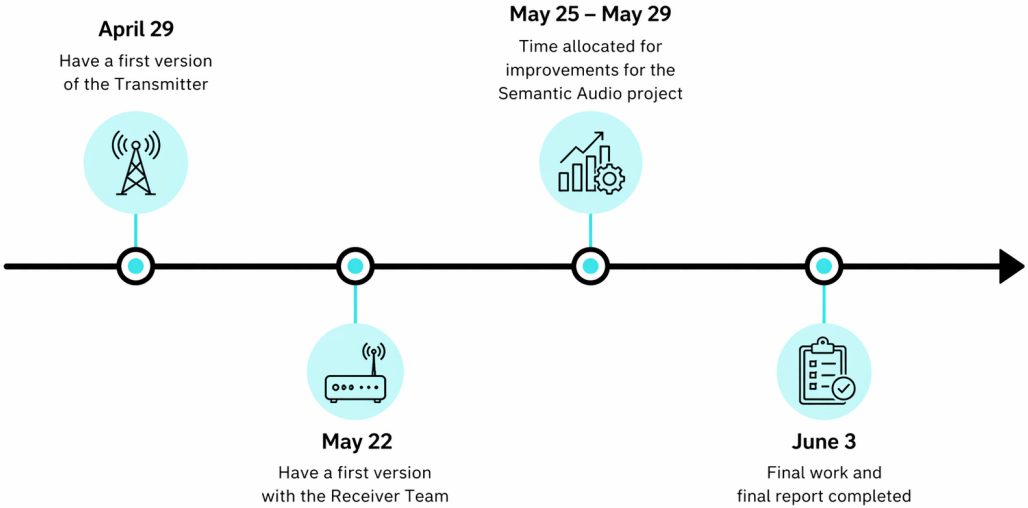
PoO \ Impact	Low	Medium	High
High			High Latency From AI Model
Medium		Insufficient Audio Quality	MQTT Communications Failures
Low		Resource Exhaustion During Demos	HW Failures

PoO - Probability of Occurence

Internal Evaluation



End of Semester Plan



References I

[1] *Guide to the Systems Engineering Body of Knowledge.*
INCOSE, 2.13 edition, 2025.

[2] adbista.
mqtt-voice-chat: A gui-based mqtt communication application.
https://github.com/adbista/mqtt-voice-chat.
Accessed: 2026-04-21.

[3] MattoMatteo.
Homegpt: Mqtt voice assistant.
https://github.com/MattoMatteo/HomeGPT.
Accessed: 2026-04-21.

[4] pinootto.
voice-mqtt: A voice interface for mqtt.
https://github.com/pinootto/voice-mqtt.
Accessed: 2026-04-21.

References II

[5] Team 9.
Deliverable 1: Analysis and requirements specification — semantic audio communication using genai – transmitter.
Technical Report Deliverable 1, Faculty of Engineering, University of Porto, March 2026.

[6] Team 9.
Deliverable 3: Market survey — semantic audio communication using genai – transmitter.
Technical Report Deliverable 3, Faculty of Engineering, University of Porto, March 2026.

[7] Team 9.
Deliverable 5: System concept — semantic audio communication using genai – transmitter.
Technical Report Deliverable 5, Faculty of Engineering, University of Porto, March 2026.

[8] Team 9.
Deliverable 6: Preliminary design — semantic audio communication using genai – transmitter.
Technical Report Deliverable 6, Faculty of Engineering, University of Porto, April 2026.

References III

[9] Team 9.
Systems Engineering Team Manual: Semantic Audio Communication using GenAI – Transmitter.
Faculty of Engineering, University of Porto, March 2026.